Problem Statement 1: Global Occupancy Rate Analysis

Objective: Calculate the total number of units, the number of currently occupied units, and the exact occupancy rate percentage across all properties combined to evaluate global portfolio utilization.

SQL Query

```
SELECT
    COUNT(Unit_ID) AS Total_Portfolio_Units,
    SUM(CASE WHEN Availability_Status = 'Occupied'
            THEN 1 ELSE 0 END) AS Occupied_Units,
    ROUND(
        (SUM(CASE WHEN Availability_Status = 'Occupied'
                 THEN 1 ELSE 0 END) * 100.0) / COUNT(Unit_ID),
        2
    ) AS Occupancy_Rate_Percentage
FROM
    Units;
```

Sample Output

Total_Portfolio_Units	Occupied_Units	Occupancy_Rate_Percentage
60	48	80.00%

Insight of Output

- **Portfolio Health:** The global portfolio is operating at a strong 80.00% occupancy rate, with 48 out of 60 total units generating active rental income.
- **Operational Buffer:** The remaining 20% (12 units) represents a mix of vacant inventory available for immediate lease on boarding and units undergoing scheduled maintenance turnovers. This leaves healthy room for portfolio scaling without sudden cash flow bottlenecks.

Problem Statement 2: Property-Wise Performance Metrics

Objective: Group data by individual property names to discover their unique total unit distribution, active tenant counts, and specific occupancy metrics. This helps identify which real estate assets are top performers and which ones suffer from under-utilization.

SQL Query

```
SELECT
    p.Property_Name,
    p.Property_Type,
    COUNT(u.Unit_ID) AS Total_Units,
    SUM(CASE WHEN u.Availability_Status = 'Occupied' THEN 1 ELSE 0
END) AS Active_Occupancies,
    SUM(CASE WHEN u.Availability_Status = 'Available' THEN 1 ELSE 0
END) AS Vacant_Units,
    SUM(CASE WHEN u.Availability_Status = 'Maintenance' THEN 1 ELSE 0
END) AS Under_Maintenance,
    ROUND(
        (SUM(CASE WHEN u.Availability_Status = 'Occupied' THEN 1 ELSE
0 END) * 100.0) / COUNT(u.Unit_ID),
        2
    ) AS Property_Occupancy_Rate_Percentage
FROM
    Properties p
LEFT JOIN
    Units u ON p.Property_ID = u.Property_ID
GROUP BY
    p.Property_ID, p.Property_Name, p.Property_Type
ORDER BY
    Property_Occupancy_Rate_Percentage DESC;
```

Sample Output

Property_ Name	Property_ Type	Total_ Units	Active_ Occupancies	Vacant_ Units	Under_ Maintenance	Property_ Occupancy_Rate _Percentage
Sayaji Heights	Commercial	15	14	1	0	93.33%
Alkapuri Residency	Apartment	20	16	3	1	80.00%
Gotri Greens	Residential	15	12	2	1	80.00%
Vasna Villas	Condo	10	6	2	2	60.00%

Insight of Output

- **Asset Champion:** Sayaji Heights (Commercial) is the top-performing asset with an exceptional 93.33% occupancy rate. Commercial setups show highly stable retention with only 1 vacant unit.
- **Underperforming Asset Risk:** Vasna Villas (Condo) stands as a critical operational bottleneck, running at a weak 60.00% occupancy rate.

Problem Statement 3: Month-on-Month (MoM) Revenue Streams

Objective: Analyze cumulative rental cash flows received on a month-by-month basis using window functions to evaluate growth trajectories and seasonal financial volatility.

SQL Query

```
WITH Monthly_Revenue AS (
    SELECT
        DATE_FORMAT(Payment_Date, '%Y-%m') AS Payment_Month,
        SUM(Amount_Paid) AS Current_Month_Revenue
    FROM
        Payments
    WHERE
        Payment_Status = 'Paid'
    GROUP BY
        DATE_FORMAT(Payment_Date, '%Y-%m')
),
Revenue_Comparison AS (
    SELECT
        Payment_Month,
        Current_Month_Revenue,
        LAG(Current_Month_Revenue, 1) OVER (ORDER BY Payment_Month) AS
Previous_Month_Revenue
    FROM
        Monthly_Revenue
)
SELECT
    Payment_Month,
    Current_Month_Revenue,
    COALESCE(Previous_Month_Revenue, 0.00) AS Previous_Month_Revenue,
    ROUND(Current_Month_Revenue - COALESCE(Previous_Month_Revenue,
Current_Month_Revenue), 2) AS Net_Change,
    CASE
        WHEN Previous_Month_Revenue IS NULL THEN '0.00%'
        ELSE CONCAT(ROUND(((Current_Month_Revenue -
Previous_Month_Revenue) / Previous_Month_Revenue) * 100.0, 2), '%')
    END AS MoM_Growth_Percentage
FROM
    Revenue_Comparison;
```

Sample Output

Payment_Month	Current_Month_Revenue	Previous_Month_Revenue	Net_Change	MoM_Growth_Percentage
2026-01	$1,12,000.00	$0.00	$0.00	0.00%
2026-02	$1,12,000.00	$1,12,000.00	$0.00	0.00%
2026-03	$1,18,500.00	$1,12,000.00	$6,500.00	5.80%
2026-04	$1,24,000.00	$1,18,500.00	$5,500.00	4.64%
2026-05	$1,14,000.00	$1,24,000.00	-$10,000.00	-8.06%
2026-06	$1,26,500.00	$1,14,000.00	$12,500.00	10.96%

Insight of Output

- **Growth Catalysts:** The portfolio saw a steady climb in March (5.80%) and April (4.64%), driven by active lease renewals at higher market premiums and successful on boarding of vacant units.
- **Seasonal Contraction Risk:** May 2026 experienced a sharp 8.06% contraction (-$10,000.00). Cross-referencing our schema rules, this drop points to localized revenue leakage either caused by an accumulation of outstanding/late tenant payments or temporary unit vacancies turning over into maintenance phases.
- **Sharp Recovery:** The business quickly bounced back in June with a 10.96% spike, proving that management recovered the delayed payments from May alongside new tenant move-ins.

Problem Statement 4: Revenue Leakage Tracking

Objective: Identify all tenants with an 'Active' lease contract whose most recent transaction is recorded as 'Late' or 'Pending'. This provides collections teams with a high-priority outreach list to eliminate cash flow leakages.

SQL Query

```
WITH RankedPayments AS (
    SELECT
        l.Lease_ID,
        l.Tenant_ID,
        l.Unit_ID,
        p.Payment_Date,
        p.Amount_Paid,
        p.Payment_Status,
        ROW_NUMBER() OVER(PARTITION BY l.Lease_ID ORDER BY
p.Payment_Date DESC) as rn
    FROM
        Leases l
    JOIN
        Payments p ON l.Lease_ID = p.Lease_ID
    WHERE
        l.Lease_Status = 'Active'
)
SELECT
    CONCAT(t.First_Name, ' ', t.Last_Name) AS Tenant_Name,
    pr.Property_Name,
    u.Unit_Number,
    u.Monthly_Rent AS Expected_Rent,
    rp.Payment_Status AS Latest_Payment_Status,
    rp.Payment_Date AS Last_Transaction_Date
FROM
    RankedPayments rp
JOIN
    Tenants t ON rp.Tenant_ID = t.Tenant_ID
JOIN
    Units u ON rp.Unit_ID = u.Unit_ID
JOIN
    Properties pr ON u.Property_ID = pr.Property_ID
WHERE
    rp.rn = 1
    AND rp.Payment_Status IN ('Late', 'Pending')
ORDER BY
    Expected_Rent DESC;
```

Sample Output

Tenant_Name	Property_Name	Unit_Number	Expected_Rent	Latest_Payment_Status	Last_Transaction_Date
Rajesh Patel	Sayaji Heights	Commercial-102	$4,500.00	Late	2026-05-05
Amit Sharma	Vasna Villas	Condo-04	$3,000.00	Pending	2026-05-10
Pooja Shah	Alkapuri Residency	Apt-302	$2,500.00	Late	2026-05-12

Insight of Output

- **Primary Exposure Core:** A total of $10,000.00 in rental revenue is currently sitting in a high-risk bucket across three active properties.
- **Commercial Risk:** The single largest contributor to this financial deficit is Rajesh Patel (Commercial-102) at Sayaji Heights, with a late payment of $4,500.00. Because commercial defaults carry larger ticket values, this profile demands immediate administrative outreach.
- **Property Correlation:** Amit Sharma (Vasna Villas) holding a pending status explains why Vasna Villas showed the lowest occupancy benchmarks in Problem 2; it is struggling with both vacant layouts and collection tracking simultaneously.

Problem Statement 5: High-Value Rent Brackets

Objective: Partition the rental catalog by asset classification (Property_Type) to isolate the top 3 highest-rent paying units using dense rank evaluations. This mapping allows management to monitor premium revenue drivers and optimize tiered asset pricing.

SQL Query

```sql
WITH Ranked_Rentals AS (
    SELECT
        p.Property_Type,
        p.Property_Name,
        u.Unit_Number,
        u.Monthly_Rent,
        DENSE_RANK() OVER (
            PARTITION BY p.Property_Type
            ORDER BY u.Monthly_Rent DESC
        ) AS Rent_Rank
    FROM
        Properties p
    JOIN
        Units u ON p.Property_ID = u.Property_ID
)
SELECT
    Property_Type,
    Property_Name,
    Unit_Number,
    Monthly_Rent,
    Rent_Rank
FROM
    Ranked_Rentals
WHERE
    Rent_Rank <= 3
ORDER BY
    Property_Type ASC,
    Rent_Rank ASC;
```

Sample Output

Property_Type	Property_Name	Unit_Number	Monthly_Rent	Rent_Rank
Apartment	Alkapuri Residency	Apt-501	$2,800.00	1
Apartment	Alkapuri Residency	Apt-302	$2,500.00	2
Apartment	Alkapuri Residency	Apt-101	$2,200.00	3
Commercial	Sayaji Heights	Commercial-201	$5,000.00	1
Commercial	Sayaji Heights	Commercial-102	$4,500.00	2
Commercial	Sayaji Heights	Commercial-105	$4,200.00	3
Condo	Vasna Villas	Condo-09	$3,500.00	1
Condo	Vasna Villas	Condo-04	$3,000.00	2
Condo	Vasna Villas	Condo-01	$2,700.00	3
Residential	Gotri Greens	Res-12	$3,200.00	1
Residential	Gotri Greens	Res-05	$3,000.00	2
Residential	Gotri Greens	Res-01	$2,800.00	3

Insight of Output

- **Premium Asset Anchors:** Commercial spaces at Sayaji Heights yield the highest baseline cash flows for the portfolio, peaking at $5,000.00/month.
- **Pricing Overlap Context:** There is an interesting premium overlap where top-tier Residential units at Gotri Greens ($3,200.00) command higher market pricing than mid-tier Condos at Vasna Villas ($3,000.00) or high-tier Apartments ($2,800.00).
- **Strategic Risk Correlation:** The data confirms that Condo-04 ($3,000.00) and Commercial-102 ($4,500.00) are top tier revenue drivers within their respective brackets. Since they were flagged with outstanding payment statuses in our leakage report, their vacancy or delinquency has a highly magnified negative impact on portfolio operations compared to standard layout turnovers.

Problem Statement 6: Maintenance Liability Outliers

Objective: Identify properties where the total estimated maintenance cost exceeds 30% of their total potential monthly rental yield (the maximum possible rent if all units were occupied). This helps management pinpoint structurally draining or inefficiently run assets.

SQL Query

```
WITH Property_Yields AS (
    SELECT
        p.Property_ID,
        p.Property_Name,
        SUM(u.Monthly_Rent) AS Potential_Monthly_Yield
    FROM
        Properties p
    JOIN
        Units u ON p.Property_ID = u.Property_ID
    GROUP BY
        p.Property_ID, p.Property_Name
),
Property_Maintenance AS (
    SELECT
        Property_ID,
        SUM(Estimated_Cost) AS Total_Maintenance_Cost
    FROM
        Maintenance_Requests
    GROUP BY
        Property_ID
)
SELECT
    py.Property_Name,
    py.Potential_Monthly_Yield,
    COALESCE(pm.Total_Maintenance_Cost, 0.00) AS
Total_Maintenance_Cost,
    ROUND(
        (COALESCE(pm.Total_Maintenance_Cost, 0.00) * 100.0) /
py.Potential_Monthly_Yield,
        2
    ) AS Cost_To_Yield_Percentage
FROM
    Property_Yields py
JOIN
    Property_Maintenance pm ON py.Property_ID = pm.Property_ID
WHERE
    (COALESCE(pm.Total_Maintenance_Cost, 0.00) * 100.0) /
py.Potential_Monthly_Yield > 30.00
ORDER BY
    Cost_To_Yield_Percentage DESC;
```

Sample Output

Property_Name	Potential_Monthly_Yield	Total_Maintenance_Cost	Cost_To_Yield_Percentage
Vasna Villas	$28,000.00	$9,800.00	35.00%

Insight of Output

- **The Structural Outlier:** Vasna Villas is the only asset breaching the critical threshold, with maintenance costs hitting 35.00% ($9,800.00) of its maximum $28,000.00 potential monthly income.
- **Operational Correlation:** This excessive maintenance overhead directly explains why the property has 2 units offline under "Maintenance" status (from Problem 2). It indicates either an aging infrastructure suffering from systemic failures or poorly managed contractor work.
- **Action Item:** Management should halt standard repairs and conduct a comprehensive structural audit of Vasna Villas to stop capital erosion.

Problem Statement 7: Lease Expiry Pipeline Risk

Objective: Identify all active leases set to expire within the next 60 days. This allows property managers to plan retention incentives or launch targeted marketing campaigns before units become vacant, protecting the portfolio's cash flow.

SQL Query

```
SELECT
    l.Lease_ID,
    p.Property_Name,
    u.Unit_Number,
    CONCAT(t.First_Name, ' ', t.Last_Name) AS Tenant_Name,
    t.Phone AS Tenant_Contact,
    l.End_Date AS Lease_Expiry_Date,
    DATEDIFF(l.End_Date, '2026-07-01') AS Days_Until_Expiry,
    u.Monthly_Rent AS Revenue_At_Risk
FROM Leases l
JOIN
    Units u ON l.Unit_ID = u.Unit_ID
JOIN
    Properties p ON u.Property_ID = p.Property_ID
JOIN
    Tenants t ON l.Tenant_ID = t.Tenant_ID
WHERE
    l.Lease_Status = 'Active'
    AND l.End_Date BETWEEN '2026-07-01' AND DATE_ADD('2026-07-01',
INTERVAL 60 DAY)
ORDER BY
    l.End_Date ASC;
```

Sample Output

Lease_ ID	Property_ Name	Unit_ Number	Tenant_ Name	Tenant_ Contact	Lease_Expiry_ Date	Days_Until_ Expiry	Revenue_ At_Risk
L-401	Gotri Greens	Res-05	Vikram Patel	+91 98765 12345	2026-07-15	14	$3,000.00
L-205	Alkapuri Residency	Apt-101	Sneha Joshi	+91 98765 54321	2026-08-05	35	$2,200.00
L-110	Sayaji Heights	Commercial-105	Rohan Mehta	+91 98765 98765	2026-08-20	50	$4,200.00

Insight of Output

- **Imminent Vacancy Risk:** The portfolio faces a total pipeline risk of $9,400.00 in monthly rental value across 3 units over the next 60 days.
- **Immediate Attention Required:** Lease L-401 (Vikram Patel) at Gotri Greens is expiring in just 14 days. If a renewal isn't signed immediately, it will create a sudden $3,000.00 monthly deficit.
- **High-Impact Commercial Target:** Rohan Mehta (Commercial-105) represents $4,200.00 of the at-risk pool. Because commercial turnovers generally take much longer to fill than apartments, the marketing team should initiate renewal conversations immediately.

Problem Statement 8: Average Turnover Velocities

Objective: Calculate the historical average duration of active lease contracts (in days) across different property types. Understanding this metric allows management to identify which asset classes offer long-term stability versus those with high turnover frequencies.

SQL Query

```
SELECT
    p.Property_Type,
    COUNT(l.Lease_ID) AS Total_Active_Leases,
    ROUND(AVG(DATEDIFF(l.End_Date, l.Start_Date)), 0) AS
Avg_Lease_Duration_Days,
    ROUND(AVG(DATEDIFF(l.End_Date, l.Start_Date)) / 30.4, 1) AS
Avg_Lease_Duration_Months
FROM
    Properties p
JOIN
    Units u ON p.Property_ID = u.Property_ID
JOIN
    Leases l ON u.Unit_ID = l.Unit_ID
WHERE
    l.Lease_Status = 'Active'
GROUP BY
    p.Property_Type
ORDER BY
    Avg_Lease_Duration_Days DESC;
```

Sample Output

Property_Type	Total_Active_Leases	Avg_Lease_Duration_Days	Avg_Lease_Duration_Months
Commercial	14	730	24.0
Residential	12	365	12.0
Apartment	16	335	11.0
Condo	6	182	6.0

Insight of Output

- **The Stability Anchor:** Commercial properties (like Sayaji Heights) show the highest tenant stability with a perfect average duration of 24 months (2 years). This long lease lifecycle lowers administrative overhead and guarantees reliable cash flow forecasting.
- **The Turnover Bottleneck:** Condos show a very short average lease lifecycle of just 6 months. Combined with our findings from Problem 2 and Problem 6 (where Vasna Villas showed the highest maintenance costs and lowest occupancy), this brief tenure implies that ongoing maintenance problems are driving tenants to move out quickly.
- **Standard Multi-Family Baseline:** Apartments and Residential units follow a predictable market standard of roughly 11 to 12 months, creating a steady, manageable rhythm for annual leasing renewals.

Problem Statement 9: Tenant Loyalty Standouts

Objective: Identify premium tenants who have maintained continuous active tenancy or contract run-times exceeding 2 consecutive years (730 days). This helps marketing teams build a VIP loyalty program to protect high-stability revenue engines.

SQL Query

```
SELECT
    t.Tenant_ID,
    CONCAT(t.First_Name, ' ', t.Last_Name) AS Tenant_Name,
    p.Property_Name,
    p.Property_Type,
    u.Unit_Number,
    l.Start_Date AS Original_Move_In,
    l.End_Date AS Current_Lease_Expiry,
    DATEDIFF(l.End_Date, l.Start_Date) AS Total_Continuous_Days,
    ROUND(DATEDIFF(l.End_Date, l.Start_Date) / 365.0, 1) AS
Continuous_Years
FROM
    Tenants t
JOIN
    Leases l ON t.Tenant_ID = l.Tenant_ID
JOIN
    Units u ON l.Unit_ID = u.Unit_ID
JOIN
    Properties p ON u.Property_ID = p.Property_ID
WHERE
    l.Lease_Status = 'Active'
    AND DATEDIFF(l.End_Date, l.Start_Date) >= 730
ORDER BY
    Total_Continuous_Days DESC;
```

Sample Output

Tenant_ID	Tenant_Name	Property_Name	Property_Type	Unit_Number	Original_Move_In	Current_Lease_Expiry	Total_Continuous_Days	Continuous_Years
T-102	Hardik Pandya	Sayaji Heights	Commercial	Commercial-201	2024-01-01	2026-12-31	1095	3.0
T-105	Deepika Vyas	Sayaji Heights	Commercial	Commercial-105	2024-06-01	2026-05-31	730	2.0
T-108	Jayesh Shah	Gotri Greens	Residential	Res-01	2024-07-01	2026-06-30	730	2.0

Insight of Output

- **The Portfolio Anchor:** Hardik Pandya (Commercial-201) stands out as the most loyal client profile, committing to a 3.0-year tenure that secures $5,000.00 in monthly premium rent.

Problem Statement 10: Chronic Property Maintenance Defects

Objective: Identify properties that have generated more than 5 critical maintenance requests within a single calendar quarter. This query filters out normal wear-and-tear to isolate systemic infrastructure failures, helping management plan major capital improvements.

SQL Query

```
SELECT
    p.Property_Name,
    p.Property_Type,
    CONCAT('Q', QUARTER(mr.Request_Date), '-', YEAR(mr.Request_Date))
AS Calendar_Quarter,
    COUNT(mr.Request_ID) AS Critical_Request_Count,
    SUM(mr.Estimated_Cost) AS Quarter_Maintenance_Cost,
    ROUND(AVG(mr.Estimated_Cost), 2) AS Avg_Cost_Per_Ticket
FROM
    Maintenance_Requests mr
JOIN
    Properties p ON mr.Property_ID = p.Property_ID
WHERE
    -- Filtering for active/severe issue tracking across a specific
quarter
    mr.Request_Date BETWEEN '2026-01-01' AND '2026-06-30'
GROUP BY
    p.Property_ID,
    p.Property_Name,
    p.Property_Type,
    YEAR(mr.Request_Date),
    QUARTER(mr.Request_Date)
HAVING
    COUNT(mr.Request_ID) > 5
ORDER BY
    Critical_Request_Count DESC;
```

Sample Output

Property_Name	Property_Type	Calendar_Quarter	Critical_Request_Count	Quarter_Maintenance_Cost	Avg_Cost_Per_Ticket
Vasna Villas	Condo	Q2-2026	7	$6,200.00	$885.71

Insight of Output

- **Systemic Infrastructure Failure:** Vasna Villas is officially flagged for chronic structural defects, logging 7 major maintenance tickets in Q2-2026 alone. This goes beyond normal tenant wear-and-tear, pointing to a core utility or structural issue (such as systemic plumbing or roof failures).
- **Financial Drag:** The cost for this single quarter's emergency tickets reached $6,200.00, making up the bulk of the property's total $9,800.00 maintenance deficit identified in Problem 6.